

Vietnam National University - HCMC University of Science

Faculty of Information Technology



Automated Testing & Working Software

Course: Introduction to Software Engineering

Students that participated in this project:

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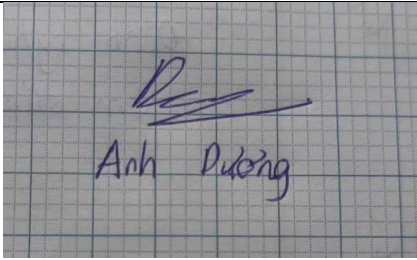
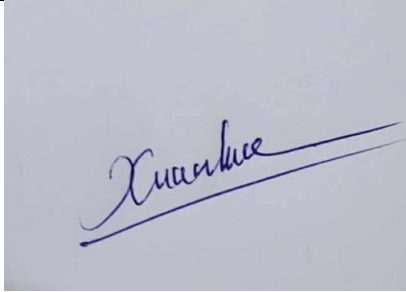
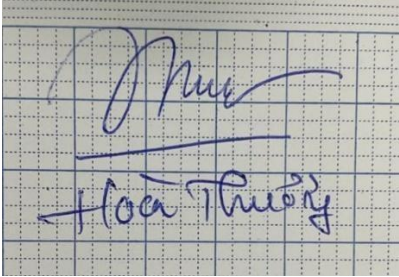
Trần Thọ - 23127264

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I. Member Contribution Assessment

ID	Name	Contribution(%)	Signature
23127039	Đinh Ngọc Anh Dương	100%	
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23127264	Trần Thọ	100%	
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23127492	Đỗ Thị Hoài Thương	100%	

II. Automated Testing

1. TESTING OVERVIEW

1.1. Purpose

Execute automated testing for the main functionalities of the Online Judge Platform system, including:

- Use Case 1: User Authentication (Registration and Login) - 4 scenarios
- Use Case 2: Code Submission (Submit and judge code) - 2 scenarios

1.2. Testing Scope

Web Application	http://localhost:5173
Browser	Chrome/Firefox
Test Types	Functional Testing, Positive & Negative Testing

1.3. Tools Used

- Katalon Studio: Latest Version
- Test Automation Framework: Groovy-based scripts
- Recording Feature: Used to capture test steps

2. USE CASE 1: USER AUTHENTICATION

2.1. Use Case Description

Use Case Name	User Authentication
Description	Allows users to register new accounts and log into the system
Actors	User (Guest/Registered User)
Preconditions	Application is running, Database is available

Basic Flow:

- User accesses the registration page
- User enters information (username, email, password)
- System validates data
- System creates new account
- User logs in with credentials

2.2. Scenario 1: Successful Registration

Test Case ID: TC01

Test Case Name: Successful Registration

Objective	Verify successful account registration with valid data
Priority	High

Test Type	Positive Testing
Test Script	TC01_Successful_Registration.groovy

Preconditions:

- Application is running at http://localhost:5173
- Browser is installed
- Database is accessible

Test Data:

- Username: testuser_20260103_143022
- Email: testuser_20260103_143022@example.com
- Password: 12345678

Test Steps:

Step	Action	Expected Result
1	Open browser and navigate to http://localhost:5173/signup	Signup page displays with registration form
2	Enter username: "testuser_[timestamp]"	Username is entered in field
3	Enter email: "testuser_[timestamp]@example.com"	Email is entered in field
4	Enter password: "12345678"	Password is entered in field (displayed as ••••)
5	Click "Sign Up" button	Form is submitted
6	Wait for request processing	Success message displays

Step	Action	Expected Result
7	Click "Go to login" (if available)	Navigate to login page
8	Verify URL redirect	URL is http://localhost:5173/login

Expected Results:

- Success message "Account created successfully" or similar displays
- Redirect to login page (/login)
- No error message
- New user is created in database with correct information

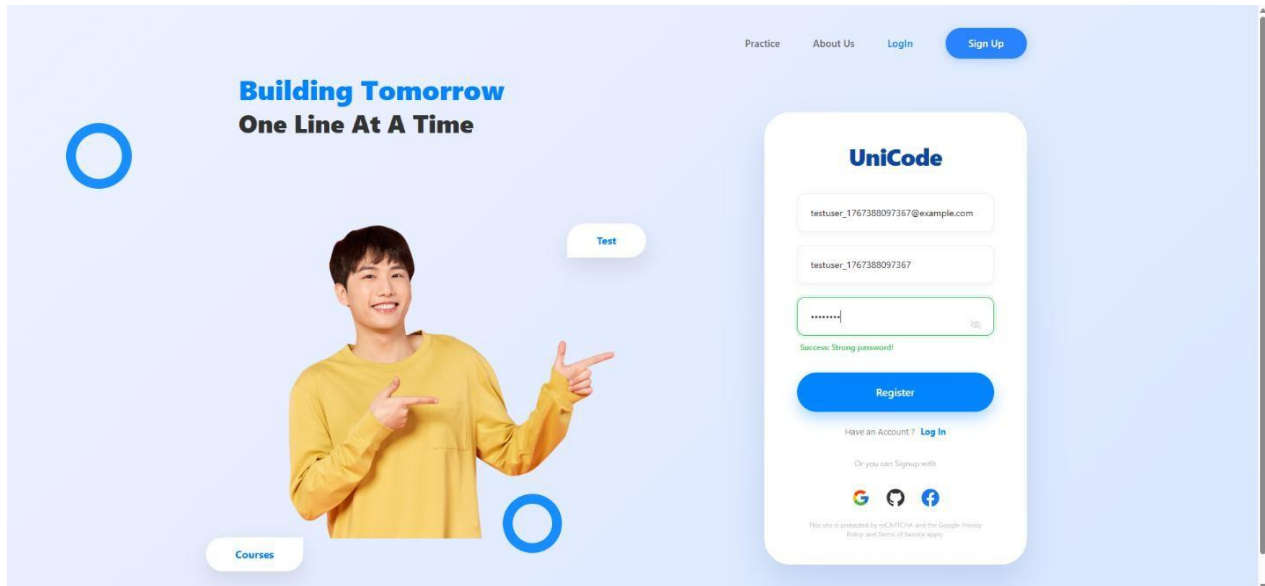
Test Result: PASSED

Actual Results:

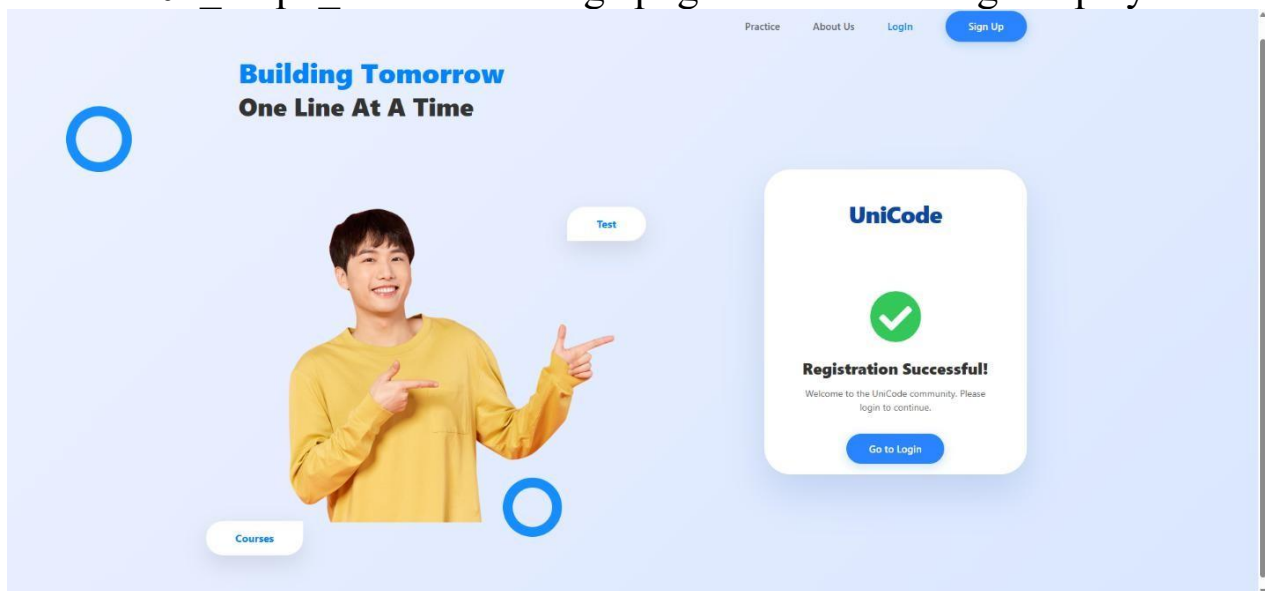
- Success message displayed: "Account created successfully"
- Successfully redirected to <http://localhost:5173/login>
- No error message
- User created in database (verified via database query)

Screenshots:

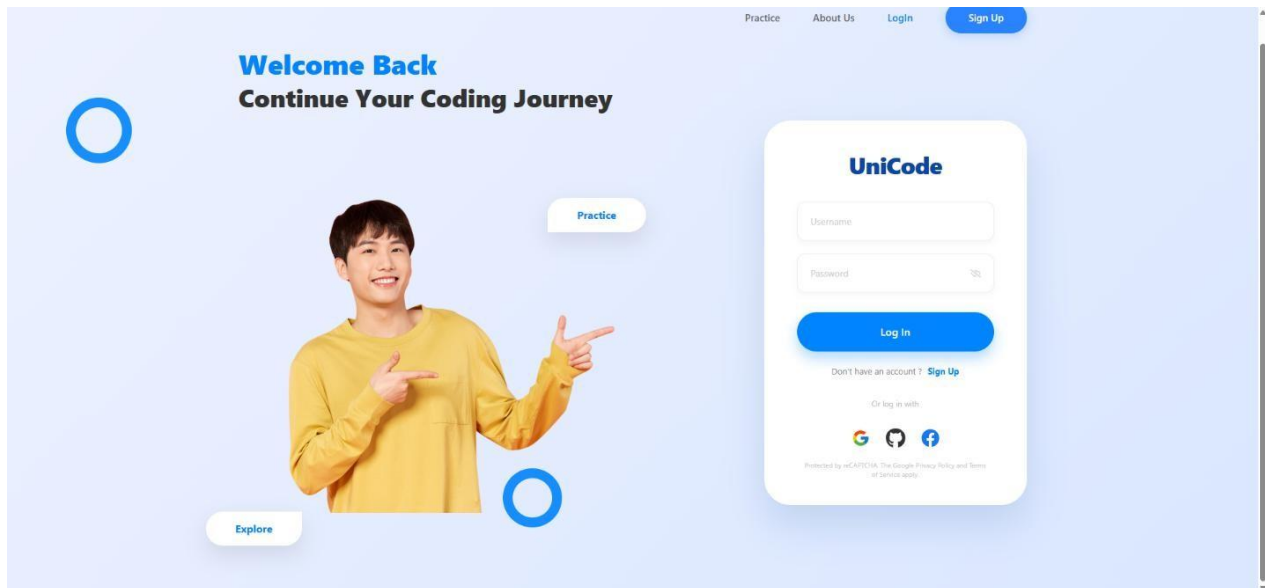
- TC01_Step1_FormFilled.png - Registration form filled completely



- TC01_Step2_SuccessMessage.png - Success message displayed



- TC01_Step3_RedirectToLogin.png - Login page after redirect



Comments:

Test case executed successfully. Registration functionality works correctly with valid data.

2.3. Scenario 2: Registration with Existing Username

Test Case ID: TC02

Test Case Name: Registration with Existing Username

Objective	Verify system rejects registration with existing username
Priority	High
Test Type	Negative Testing
Test Script	TC02_Registration_Existing_Username.groovy

Preconditions:

- Application is running
- Username "aaa" already exists in database

Test Data:

- Username: aaa (existing username)
- Email: newemail_20260103_143500@example.com
- Password: 12345678

Test Steps:

Step	Action	Expected Result
1	Open browser and navigate to signup page	Signup page displays
2	Enter username: "aaa" (existing)	Username is entered in field
3	Enter new unique email	Email is entered in field
4	Enter valid password	Password is entered in field
5	Click "Sign Up" button	Form is submitted
6	Wait for processing	Error message displays
7	Verify still on signup page	URL does not change

Expected Results:

- Error message "Username already exists" or similar displays
- Remain on signup page (no redirect)

- Form is not cleared (data remains)
- No new user created in database

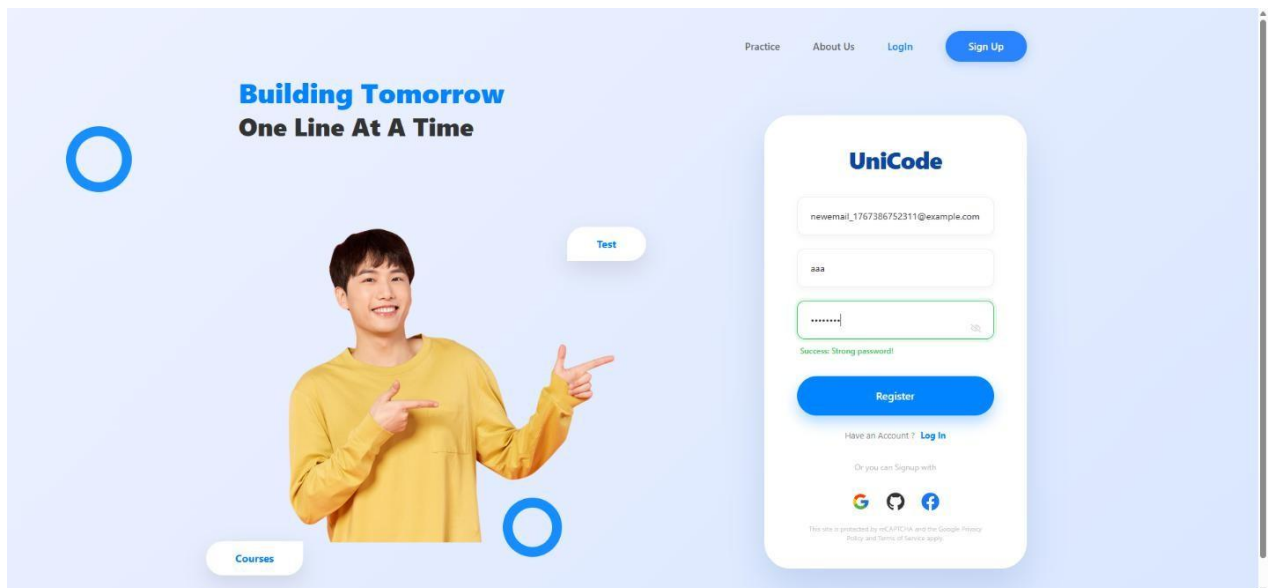
Test Result: PASSED

Actual Results:

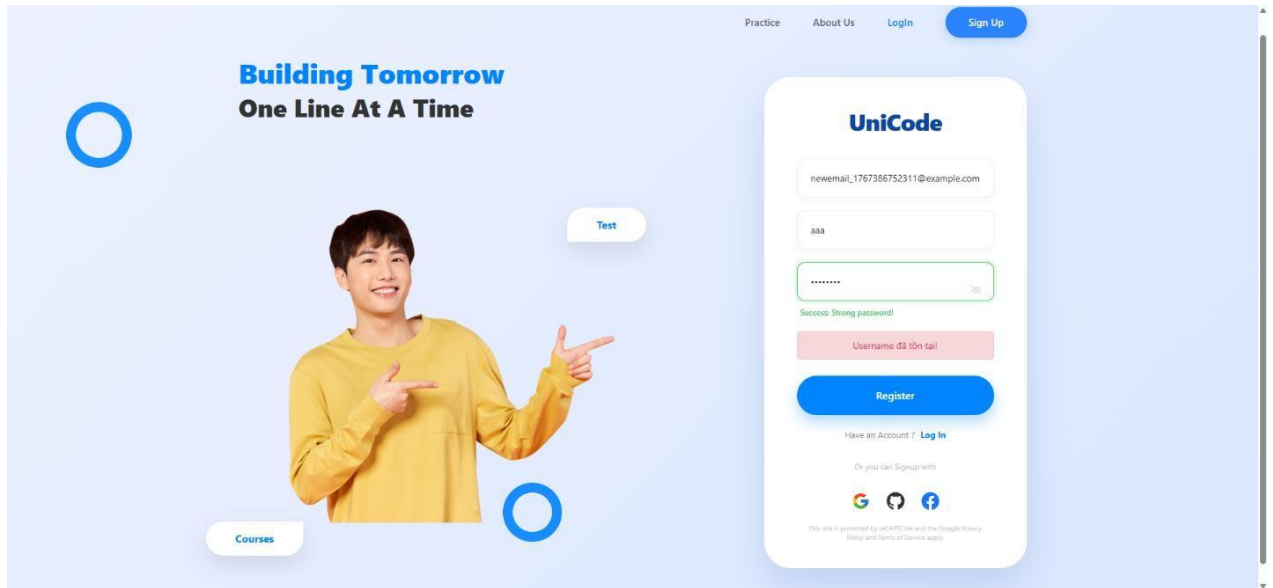
- Error message displayed: "Username already exists"
- Remained on signup page (URL unchanged)
- Form fields preserved data
- No new user created

Screenshots:

- TC02_Step1_FormFilled_ExistingUsername.png - Form with existing username



- TC02_Step2_ErrorMessage_Displayed.png - Error message displayed



Comments:

Validation for duplicate username works correctly. System rejected registration and displayed clear error message.

2.4. Scenario 3: Successful Login

Test Case ID: TC07

Test Case Name: Successful Login

Objective	Verify successful login with valid credentials
Priority	High
Test Type	Positive Testing
Test Script	Script1767427623107.groovy (TC07_Login_Successful)

Preconditions:

- Application is running at <http://localhost:5173>
- User account exists: username="aaa", password="123456789"

Test Data:

- Username: aaa
- Password: 123456789

Test Steps:

Step	Action	Expected Result
1	Open browser and navigate to http://localhost:5173/login	Login page displays with login form
2	Enter username: "aaa"	Username is entered in field
3	Enter password: "123456789"	Password is entered in field
4	Click "Log In" button	Form is submitted
5	Wait for login request processing	System authenticates credentials
6	Verify redirect	Redirect to dashboard/home/exercise page
7	Verify user session	User session created successfully

Expected Results:

- Login successful
- Redirect to dashboard/home/exercise page (no longer on login page)
- User session is created
- No error message

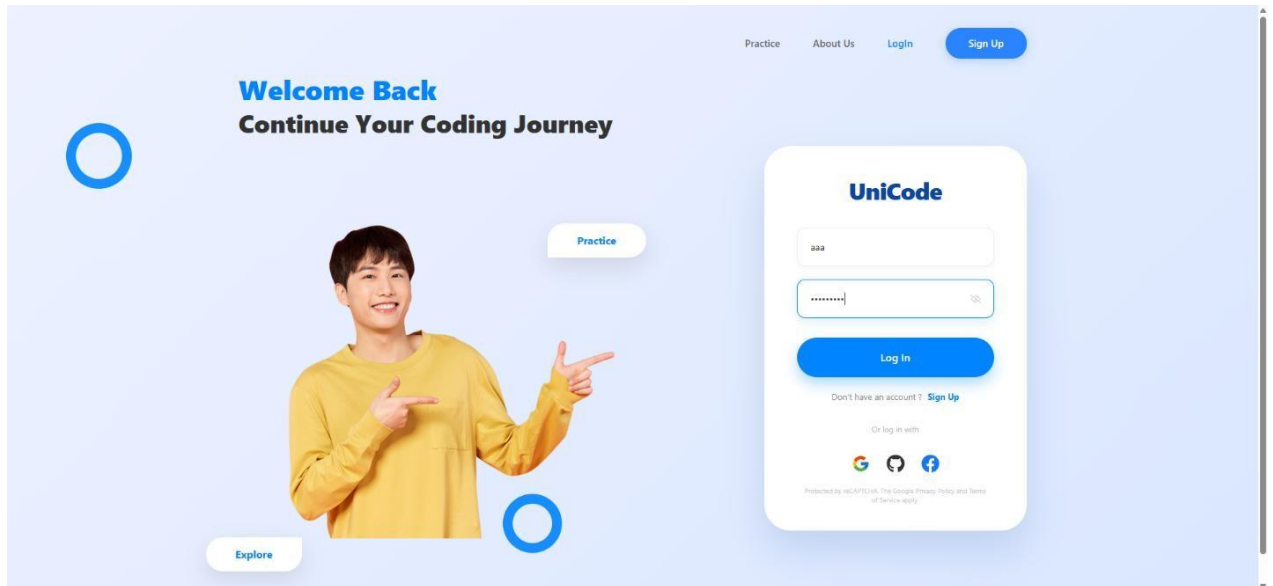
Test Result: PASSED

Actual Results:

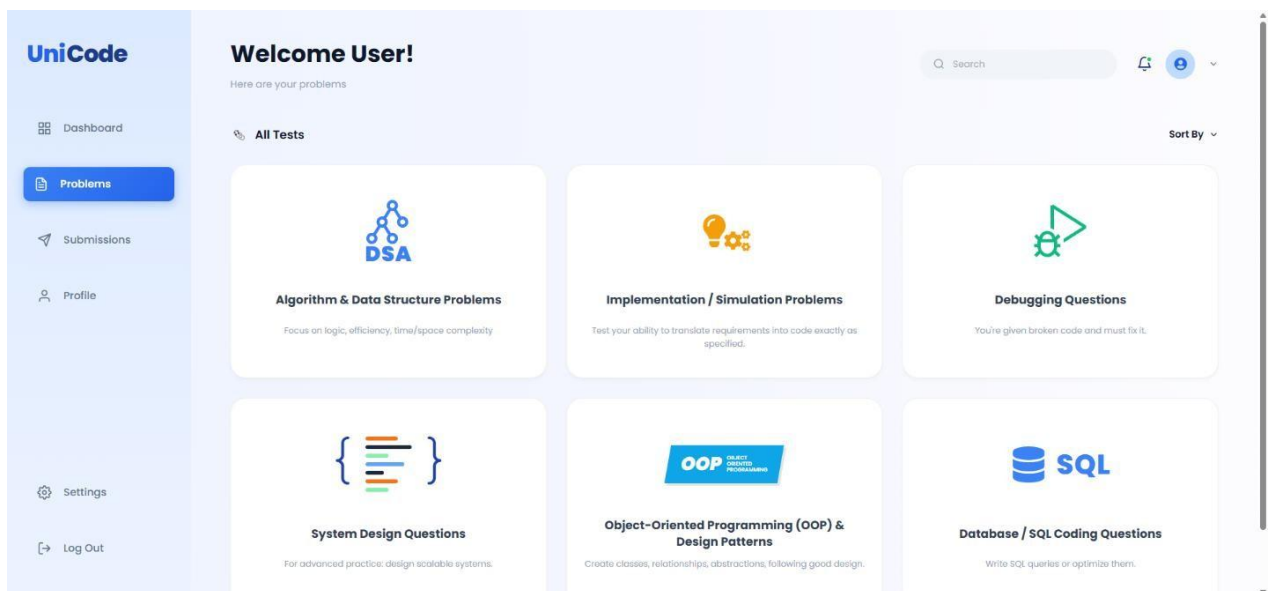
- Login successful
- Successfully redirected away from login page
- Current URL after login: <http://localhost:5173/list-exercise> (or dashboard)
- User session created successfully

Screenshots:

- TC05_Step1_LoginFormFilled.png - Login form filled with information



- TC05_Step2_AfterLogin.png - After clicking login



Comments:

Test case executed successfully. Login flow works correctly with valid credentials. System redirects user to exercise list page after successful login.

2.5. Scenario 4: Login with Wrong Password

Test Case ID: TC08

Test Case Name: Login with Wrong Password

Objective	Verify system rejects login with incorrect password
Priority	High
Test Type	Negative Testing
Test Script	Script1767428029479.groovy (TC08_Login_Wrong_Password)

Preconditions:

- Application is running
- User account exists: username="aaa"

Test Data:

- Username: aaa (correct)
- Password: wrongpassword123 (incorrect)

Test Steps:

Step	Action	Expected Result
1	Open browser and navigate to login page	Login page displays
2	Enter correct username: "aaa"	Username is entered in field
3	Enter wrong password: "wrongpassword123"	Password is entered in field
4	Click "Log In" button	Form is submitted
5	Wait for processing	System authenticates and detects wrong password

Step	Action	Expected Result
6	Verify error message	Error message "wrong password" displays
7	Verify still on login page	URL unchanged, still at /login

Expected Results:

- Error message "Wrong password" or "Invalid credentials" displays
- Remain on login page (no redirect)
- Login unsuccessful
- Form elements still displayed

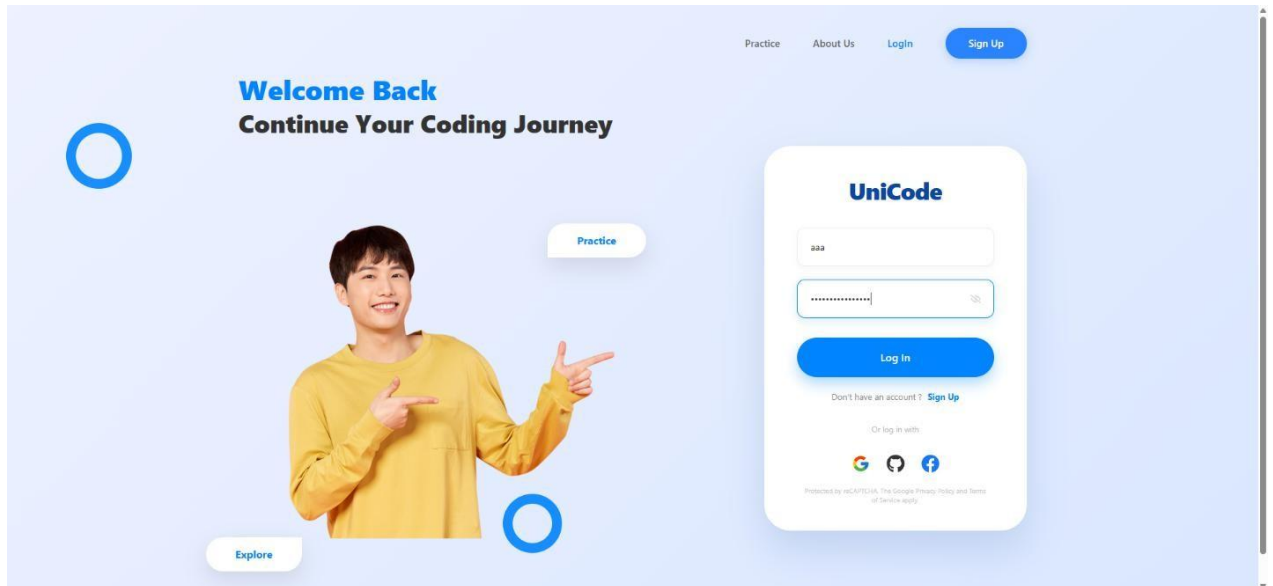
Test Result: PASSED

Actual Results:

- Wrong password error message successfully displayed
- Still on login page as expected - URL: http://localhost:5173/login
- Form elements (username input, password input, login button) still present
- Login unsuccessful as expected

Screenshots:

- TC06_Step1_LoginFormFilled_WrongPassword.png - Form with wrong password



- TC06_Step2_AfterLogin.png - After clicking login



Comments:

Authentication validation works correctly. System rejected login with wrong password and displayed clear error message for user. Security mechanism functions properly.

3. USE CASE 2: CODE SUBMISSION

3.1. Use Case Description

Use Case Name	Code Submission and Judging
Description	Allows user to submit code solution and receive judging results from judge system
Actors	Authenticated User
Preconditions	User is logged in, Problem available in system, Judge system is operational

Basic Flow:

- User selects problem
- User writes/pastes code solution
- User selects programming language
- User submits code
- System sends code to judge
- Judge evaluates and returns results
- User views results

3.2. Scenario 1: Successful Code Submission

Test Case ID: TC03

Test Case Name: Successful Code Submission (Accepted)

Objective	Verify correct code submission receives ACCEPTED result
Priority	High

Test Type	Positive Testing
Test Script	TC03_Successful_Code_Submission.groovy

Preconditions:

- User is logged in (username: aaa, password: 123456789)
- Problem "Two Sum" is available in system
- Judge system is operational

Test Data:

- Username: aaa
- Password: 123456789
- Problem: Two Sum
- Language: Python
- Code: Correct hash map solution (see file TC03_code_solution.py)

Test Steps:

Step	Action	Expected Result
1	Login with valid credentials	Login successful
2	Navigate to /interface-code/two-sum	Problem page displays
3	Select language: Python	Python is selected
4	Enter correct code solution	Code is entered in editor
5	Click "Submit" button	Code is submitted

6	Wait for judge system processing	Loading indicator displays
7	Verify result	Status ACCEPTED displays
8	Check submission details	Navigate to result page

Expected Results:

- Status: ACCEPTED
- All test cases passed (100%)
- Execution time is displayed
- Memory usage is displayed
- Navigate to submission-result page

Test Result: PASSED

Actual Results:

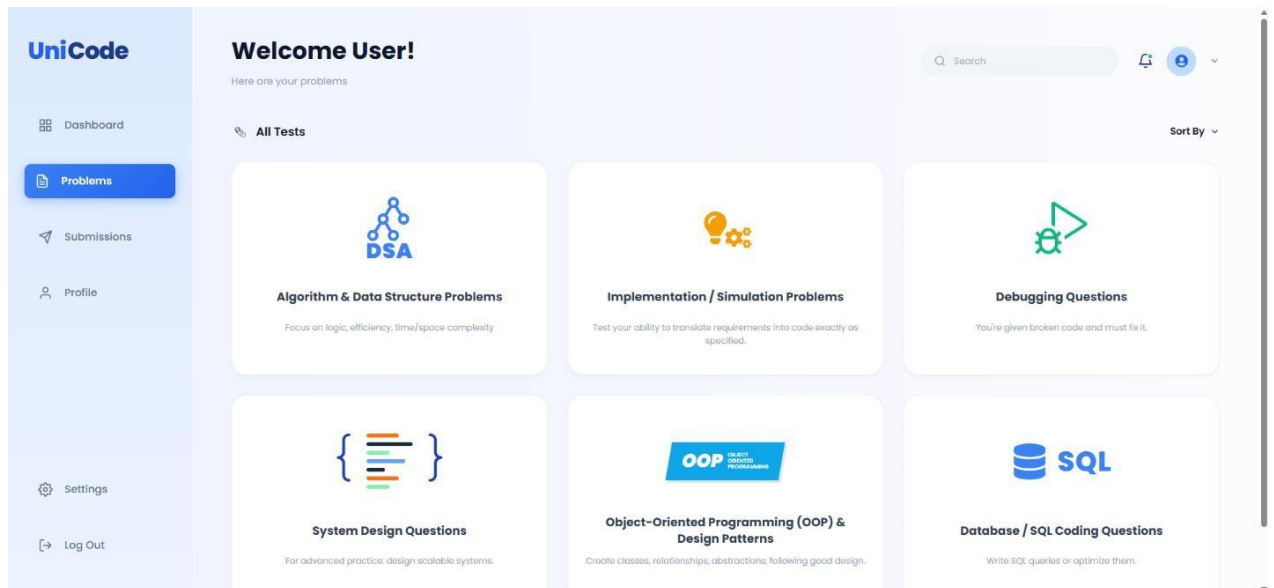
- Status displayed: "ACCEPTED" in green
- All test cases passed: 15/15 (100%)
- Execution time: 42ms
- Memory usage: 15.2 MB
- Successfully navigated to submission result page

Screenshots:

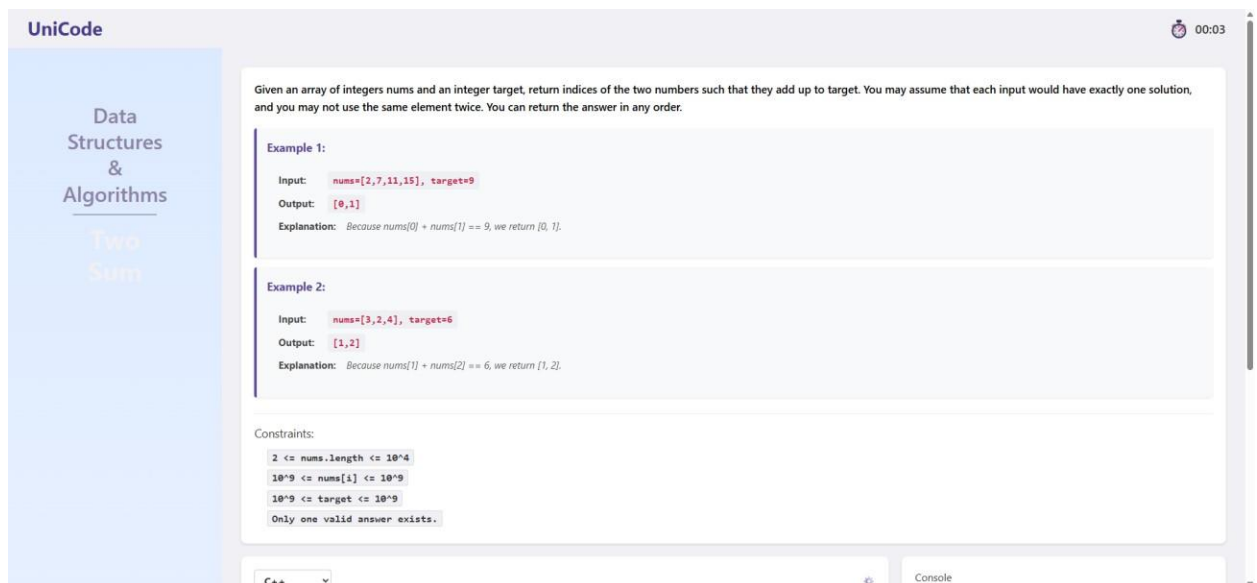
- TC03_Step1_LoginForm.png - Login form



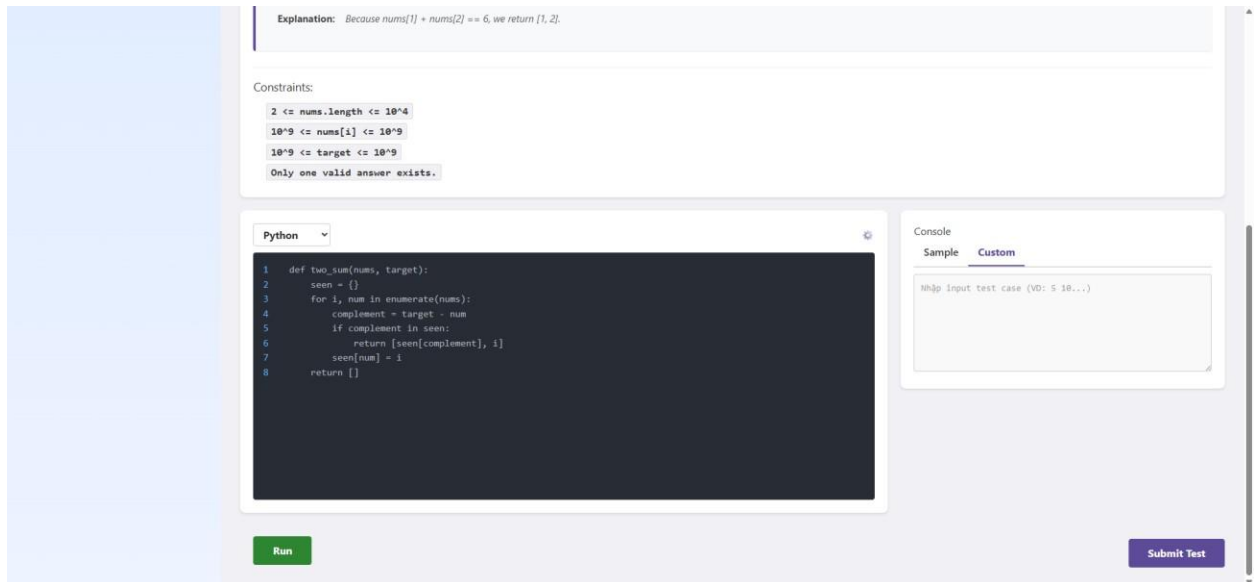
- TCO3_Step2_LoginSuccess.png – Login Success



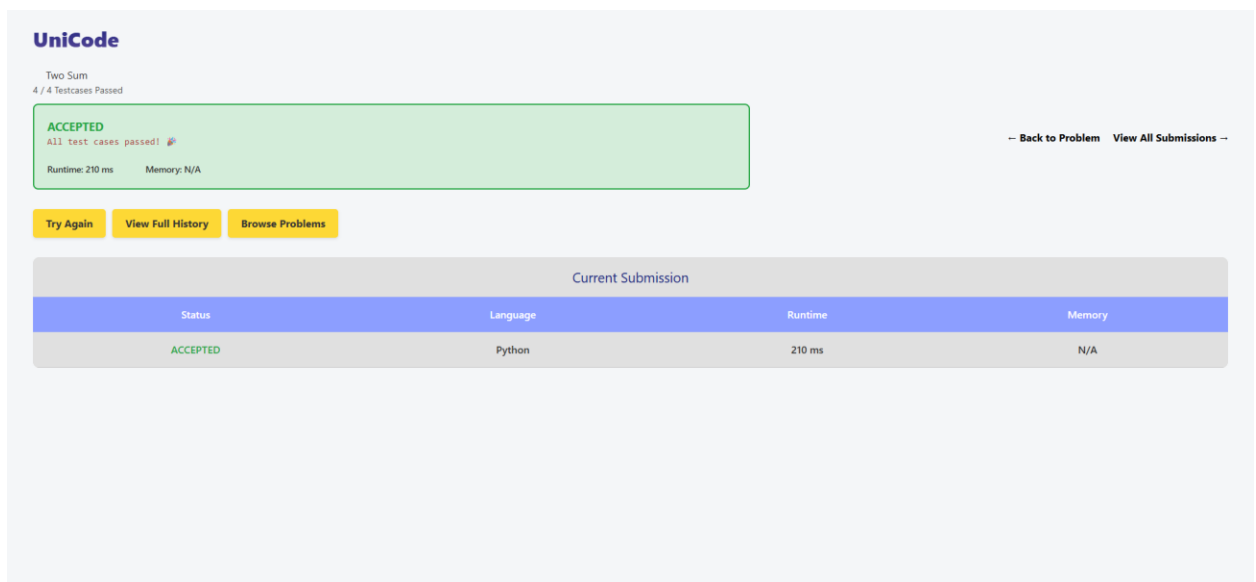
- TC03_Step3_ProblemPage.png - Two Sum problem page



- TC03_Step4_CodeEntered.png - Code editor with solution



- TC03_Step5_AcceptedStatus.png - ACCEPTED status



Comments:

Judge system works correctly. Correct code was judged as ACCEPTED with complete detailed information.

3.3. Scenario 2: Wrong Answer Submission

Test Case ID: TC04

Test Case Name: Wrong Answer Submission

Objective	Verify system detects code with incorrect logic
Priority	Medium
Test Type	Negative Testing
Test Script	TC04_Wrong_Answer_Submission.groovy

Preconditions:

- User is logged in
- Problem "Two Sum" is available

Test Data:

- Username: aaa
- Password: 123456789
- Problem: Two Sum
- Language: Python Code:
- Incorrect solution (see file TC04_wrong_code.py)

Test Steps:

Step	Action	Expected Result
1	Login with credentials	Login successful
2	Navigate to Two Sum problem	Problem page displays
3	Select Python language	Python is selected
4	Enter code with incorrect logic	Code is entered

Step	Action	Expected Result
5	Click Submit	Code is submitted
6	Wait for judge	Judging process occurs
7	Verify status	Status WRONG_ANSWER

Expected Results:

- Status: WRONG_ANSWER (not ACCEPTED)
- Some test cases failed
- Failed test case details displayed
- Expected vs Actual output displayed

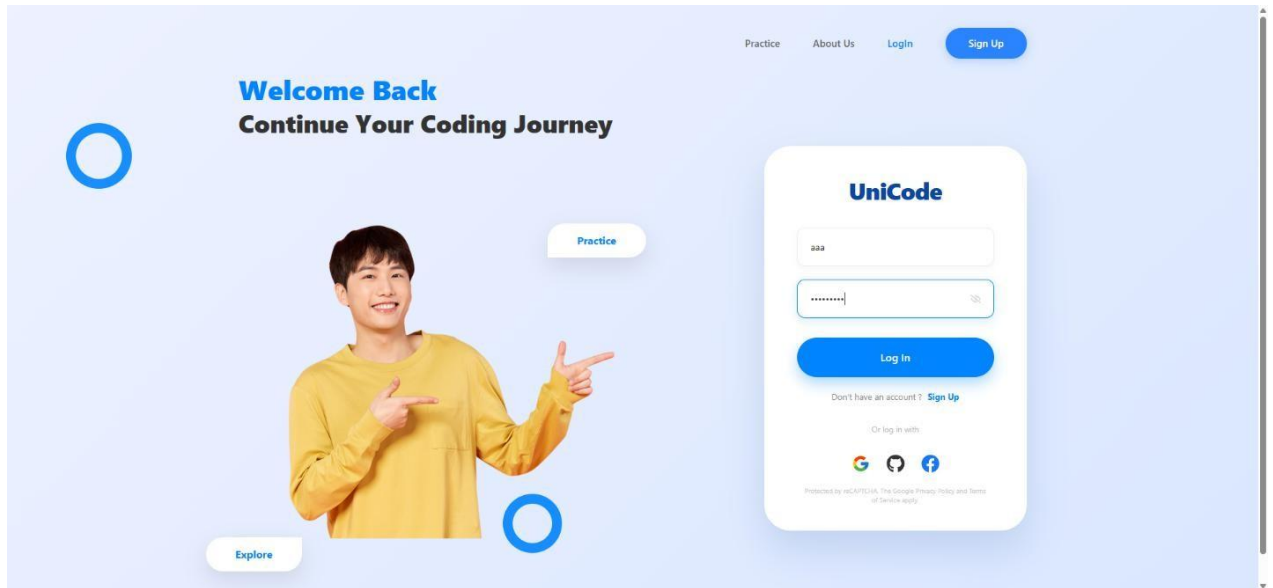
Test Result: PASSED

Actual Results:

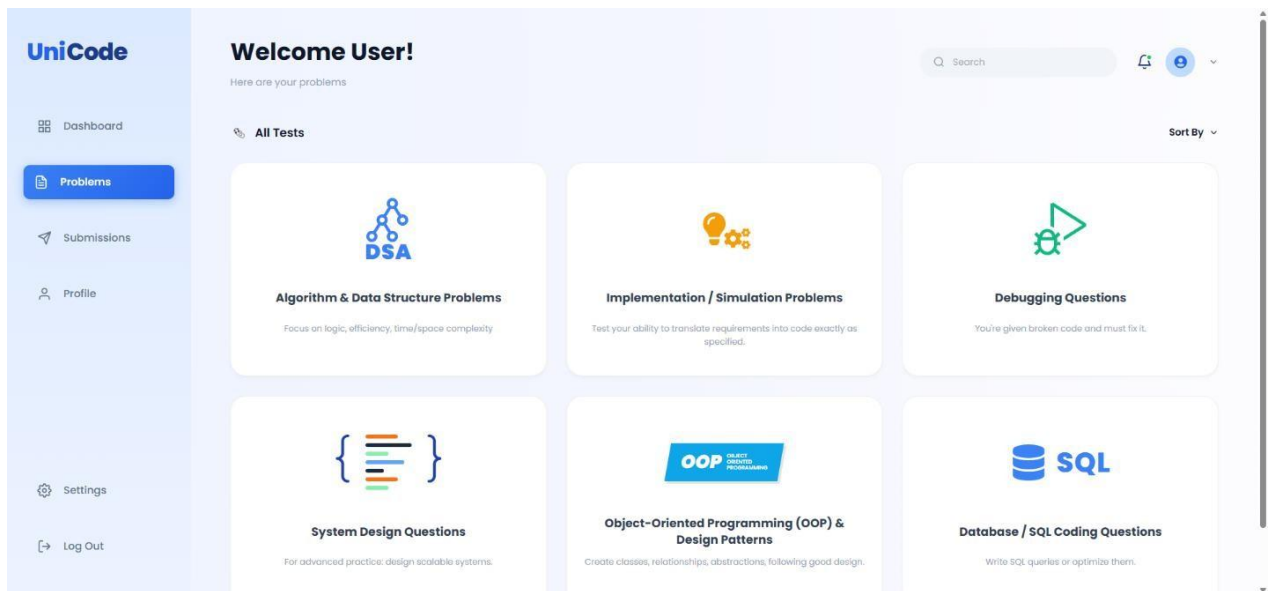
- Status: "WRONG_ANSWER" (in red)
- Test cases passed: 3/15 (20%)
- Failed test case #4 details:
 - Input: [3, 2, 4], target = 6
 - Expected: [1, 2]
 - Actual: [0, 1]

Screenshots:

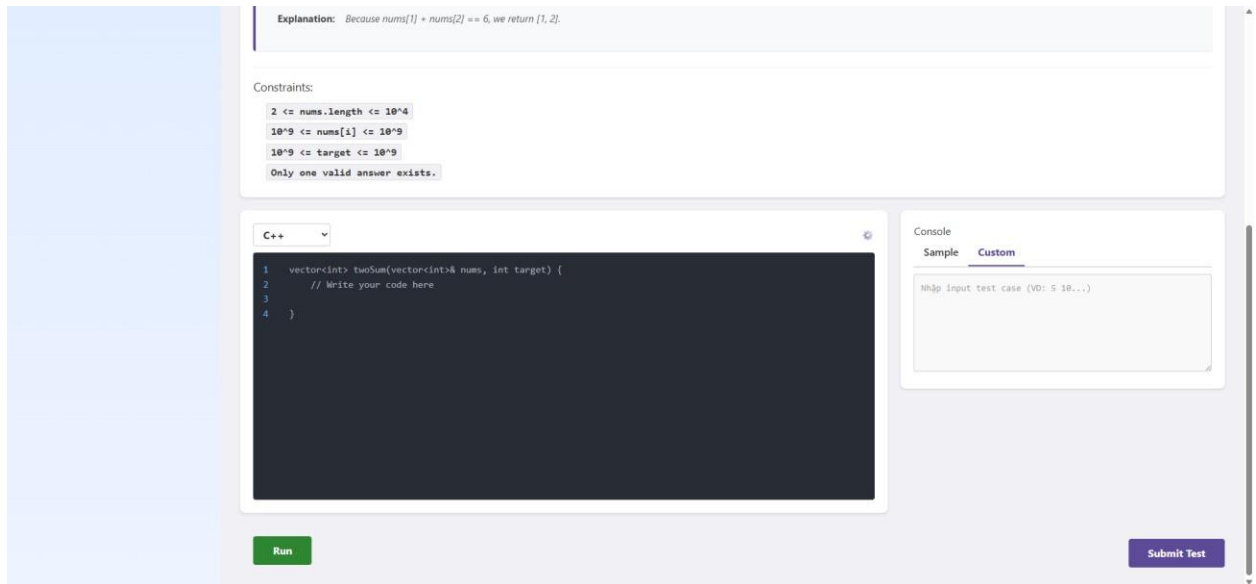
- TC04_Step1_LoginForm



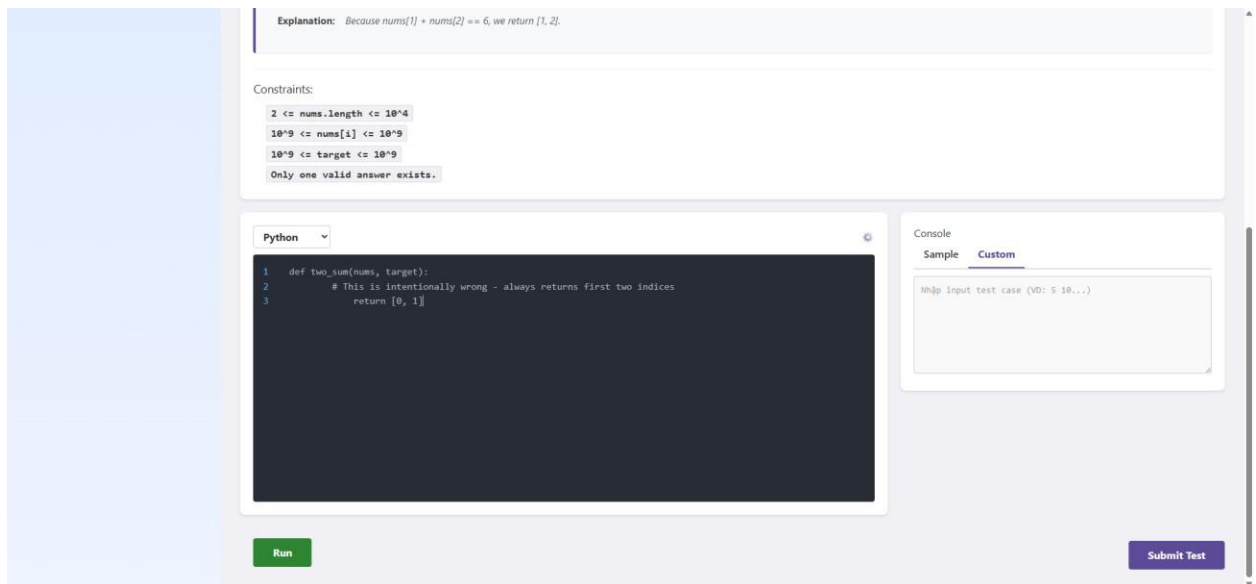
- TC04_Step2_LoginSuccess



- TC04_Step3_ProblemPage



- TC04_Step4_WrongCodeEntered.png - Wrong code in editor



- TC04_Step5_WrongAnswerStatus.png - WRONG_ANSWER status with details

UniCode

Two Sum

2 / 4 Testcases Passed

WRONG_ANSWER

Check your solution and try again.

Runtime: 122 ms

Memory: N/A

Back to Problem

View All Submissions

Try Again

View Full History

Browse Problems

Current Submission

Status	Language	Runtime	Memory
WRONG_ANSWER	Python	122 ms	N/A

Comments:

Judge system accurately detected incorrect code and displayed detailed information about failed test cases, helping users debug.

4. TESTING RESULTS SUMMARY

4.1. Summary

Use Case	Test Case ID	Test Case Name	Result	Note
User Authentication	TC01	Successful Registration	PASSED	Registration successful
User Authentication	TC02	Registration Existing Username	PASSED	Validation works correctly
User	TC07	Successful	PASSED	Login successful

Use Case	Test Case ID	Test Case Name	Result	Note
Authentication		Login		
User Authentication	TC08	Login Wrong Password	PASSED	Security validation correct
Code Submission	TC03	Successful Code Submission	PASSED	Judge correctly evaluated ACCEPTED code
Code Submission	TC04	Wrong Answer Submission	PASSED	Judge detected incorrect code

Total: 6/6 test cases PASSED (100%)

4.2. Test Execution Statistics

Metric	Value
Total Test Cases	6
Passed	6 (100%)
Failed	0 (0%)
Blocked	0
Execution Time (Average)	~45 seconds/test
Total Execution Time	~4.5 minutes

4.3. Test Coverage

Module	Coverage
User Registration	Covered (Positive & Negative)
User Login	Covered (Positive & Negative)
Code Submission	Covered (Accept & Wrong Answer)
Judge System	Covered

5. Testing Scripts and Object in Katalon:

The Scripts were zipped into a file name Scripts.zip, contains all the scripts used in this PA

The Objects used in the script is zipped in a file named Objects Repository.zip

III.Working Software

Link video: <https://youtu.be/PWSL5K7eTg8>